📘 Standard Operating Procedures (SOPs) in Computer Forensics

SOPs are **documented procedures** that define how specific digital forensic tasks must be performed to ensure:

- **Consistency**
- **Integrity of evidence**
- **Legal compliance**
- **Auditability**

🔐 Why SOPs are Critical:

- Prevent **contamination or loss** of evidence

- Ensure **repeatability** of results
- Help pass **legal scrutiny** in court
- Guide even in **high-stress, real-time incidents**

Core Elements of a Forensic SOP:

Step	Description
Preparation	Ensure tools, team, legal authority, and documentation are ready
First Response	Secure the scene and systems
Preservation	Isolate the system and create forensic images
Examination	Analyze systems using proper tools (non-invasive)
Documentation	Record everything (tools used, hashes, timeline, etc.)
Reporting	Generate a formal report detailing findings
Review	Internal review and continuous improvement

Processing Crime and Incident Scenes – Windows & DOS Systems

Investigators often work with **legacy systems** (DOS) and **modern systems** (Windows XP to Windows 11). Here's how to approach them:

1. Before Touching the System:

- Photograph the system and surrounding environment
- Document all **visible information** (e.g., screen contents)
- Record **connected peripherals** (USBs, network cables)

- Note **powered-on status**, time displayed, background applications

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2. Working with Powered-On Systems:

Especially on **Windows** or **DOS**, powered-on systems must be treated carefully:

Step	Windows	DOS
Capture RAM	Use tools like FTK Imager Lite, Belkasoft RAM Capturer	Not applicable
Get running processes	tasklist, wmic, or tools like Volatility later	Limited to TSR programs
Open ports/network info	netstat, ipconfig, TCPView	No networking in DOS
Dump registry	Use reg save	DOS doesn't have registry
Screenshot current screen	Snipping tool, nircmd	Use camera for DOS

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3. Shut Down or Not?

Best practice: **Preserve live system memory** before powering down.

- If encryption or malware is active, **shutting down may destroy evidence**.

If forced to power off:

- Perform **forensic imaging** immediately
- Avoid **normal shutdown**; unplug power directly (forensic decision)

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4. Imaging Disks (Preservation):

Use **write blockers** and tools like:

- FTK Imager
- Guymager
- dd (for DOS, via bootable forensic OS)
- EnCase

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5. Examining Files:

- In Windows, analyze:
  - C:\Users\ profiles
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    - AppData, Prefetch, Recent, Recycle Bin
    - Windows Event Logs (.evtx)
    - Registry hives: NTUSER.DAT, SAM, SYSTEM
- In DOS:
    - Check .BAT files, autoexec.bat, config.sys
    - Look for custom scripts or resident programs (.COM, .EXE)

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6. Special Considerations for DOS Systems:

DOS systems are rare but may still be used in **industrial control systems (ICS), ATMs**, or legacy hardware.

Characteristic	Impact
No multitasking	No background logging
No registry	Fewer artifacts
File system	FAT16/FAT32
Log files	Very limited

For such systems, **disk-level analysis is critical**.

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7. Reporting & Chain of Custody

- Record:
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Exact time of interaction

- Every command or tool used

- Hashes of collected evidence

- Use proper **Evidence Forms** and **Chain of Custody Logs**

🛠 Tools for Windows and DOS Forensics:

Tool	Platform	Use
FTK Imager	Windows	Imaging, preview
Volatility	Windows memory	RAM analysis
Autopsy/Sleuth Kit	Cross-platform	File, metadata, timeline
DOSBox + Disk Editors	DOS	Simulate/analyze DOS apps
WinHex	Both	Hex editing, forensic imaging

Summary: Best Practices for Windows/DOS Scene Processing

- Minimize interaction — **don’t alter anything unless necessary**
- Always use **write-blockers**
- If in doubt, **clone first, analyze later**
- For DOS: focus on **disk-level** evidence and **manual logs**
- For Windows: look at **registry, event logs, prefetch**, and **RAM dump**